

*Answers must be given **exclusively on this sheet**: answers given on the other sheets **will be ignored**. This question paper has **6 questions** and **2 pages**.*

NAME AND ROLL NUMBER

Please mark your Roll No:

☐0 ☐0 ☐0 ☐0 ☐0 ☐0 ☐0 ☐0 ☐0  
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Name of the Student:

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Roll Number of the Student:

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Signature of the Student:

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Signature of the Invigilator:

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QUESTIONS AND RESPONSES

**Question [q1]:** (2 points) Let a linear model  $Y = \beta_0 + \beta_1 X + \epsilon$  be fitted to the following data, where  $\epsilon$  is normally distributed with mean 0 and unknown variance  $\sigma^2 > 0$ .

|       |   |   |   |   |   |
|-------|---|---|---|---|---|
| $x_i$ | 0 | 1 | 2 | 3 | 4 |
| $y_i$ | 3 | 4 | 5 | 6 | 7 |

Let  $\hat{Y}_0$  denote the ordinary least-square estimator of  $Y$  at  $X = 6$ . If  $E(\hat{Y}) = c_0\beta_0 + c_1\beta_1$  and  $Var(\hat{Y}_0) = d\sigma^2$ , then which of the following is/are true?

- ☐  $c_1 = 6, c_2 = 1, 1.7 \leq d \leq 1.9$ 
☒  $c_1 + c_2 + d \leq 8.8$   
☒  $c_1 = 1, c_2 = 6, 1.7 \leq d \leq 1.9$ 
☒  $3.1 \leq c_2 - c_1 - d \leq 3.5$

**Question [q2]:** (3 points) Let  $y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \cdots + \beta_{22} x_{22,i} + \epsilon_i, i = 1, 2, \dots, 123$ , where,  $x_{1,i}, \dots, x_{22,i}$  are fixed input (independent) variables; and, for  $j = 0, 1, 2, \dots, 22$ ;  $\beta_j$  are unknown parameters, and  $\epsilon_i$ 's are independent and identically distributed normal random variables with mean zero and finite variance. If  $SS_{Reg} = 338.92$  and  $SS_T = 522.30$  and the value of  $R_{adj}^2$  (adjusted  $R^2$ ) equals to  $t$ . Then which of the following statements is/are true?

- ☐  $0.61 < t < 0.69$ 
☒  $0.51 < t < 0.59$   
☐  $0.43 < t < 0.54$ 
☒  $0.53 \leq t < 0.63$

**Question [q3]:** (2 points) Consider a multiple linear regression problem. Which of the following statements is/are true?

- ☐ A leverage point is always influential point  
☒ PRESS is a measure of how well a regression model perform in predicting new observations  
☒ Q-Q plot is used to test for normality  
☒ A linear model is linear in the parameters, but not necessarily in the independent variables

**Question [q4]:** (3 points) Consider the multiple linear regression problem

$$Y = X\beta + \epsilon, \text{ and } \hat{Y} = X\hat{\beta}$$

where  $Y = (Y_1, \dots, Y_n)'$  is a  $n \times 1$  vector of response variable,  $X$  is a  $n \times p$  design matrix with full rank (largest possible rank),  $\beta$  is a  $p \times 1$  vector of parameters, and  $\epsilon$  is a  $n \times 1$  vector of errors;  $n > p$ . The first element of  $\beta$  is the intercept of the model. Then which of the following statements is/are true?

☐  $\sum_{i=1}^n \text{Var}(\hat{Y}_i) = \sigma^2 n$

☒  $\sum_{i=1}^n \text{Var}(\hat{Y}_i) < \sigma^2 n$

☐  $\sum_{i=1}^n \text{Var}(\hat{Y}_i) < \sigma^2 p$

☒  $\sum_{i=1}^n \text{Var}(\hat{Y}_i) = \sigma^2 p$

**Question [q5]:** (3 points) For  $Y \in \mathbb{R}^n$ ,  $X \in \mathbb{R}^{n \times p}$ , and  $\beta \in \mathbb{R}^p$ , consider a regression model

$$Y = X\beta + \epsilon,$$

where  $\epsilon$  has an  $n$ -dimensional multivariate normal distribution with zero mean vector and identity covariance matrix. Let  $I_p$  denote the identity matrix of order  $p$ . For  $\lambda > 0$ , let

$$\hat{\beta}_n = (X^T X + \lambda I_p)^{-1} X^T Y,$$

be an estimator of  $\beta$ . Then which of the following options is/are correct?

☐  $\hat{\beta}_n$  is an unbiased estimator of  $\beta$

☒  $(X^T X + \lambda I_p)$  is a positive definite matrix

☒  $\hat{\beta}_n$  has a multivariate normal distribution

☐  $\text{Var}(\hat{\beta}_n) = (X^T X + \lambda I_p)^{-1}$

**Question [q6]:** (2 points) Consider the linear model

$$Y_1 = \beta_1 + \beta_2 + \epsilon_1$$

$$Y_2 = \beta_1 + 2\beta_2 + \epsilon_2$$

$$Y_3 = \beta_1 + c\beta_2 + \epsilon_3,$$

where  $\beta_1, \beta_2 \in \mathbb{R}$  are unknown parameters, and the uncorrelated errors  $\epsilon_i, i = 1, 2, 3$  have zero mean and finite variance  $\sigma^2 (> 0)$ . The constant  $c$  is such that  $\hat{\beta}_1$  and  $\hat{\beta}_2$  are uncorrelated, where  $\hat{\beta}_1$  and  $\hat{\beta}_2$  are the best linear unbiased estimators of  $\beta_1$  and  $\beta_2$ , respectively. Then  $(\text{Var}(\hat{\beta}_1), \text{Var}(\hat{\beta}_2))$  equals

☒  $(\frac{\sigma^2}{3}, \frac{\sigma^2}{14})$

☐  $(3\sigma^2, \frac{\sigma^2}{14})$

☐  $(\frac{\sigma^2}{14}, \frac{\sigma^2}{3})$

☐  $(\frac{\sigma^2}{3}, 14\sigma^2)$